STA 322 Study Design Project 1
Team Members: Eric Guan & Kevin Liu

Introduction

In this project, we aim to learn more about the characteristics of Duke faculty in Trinity College of Arts & Sciences. More specifically, our main goals are to estimate the fraction of faculty who are women, the fraction who have at least one grant ongoing in 2026, and, among faculty who have a Ph.D., the average number of years since they received their degree and the fraction who earned their Ph.D. at Duke University. Instead of collecting outcome data on the entire population, we use a probability sample to estimate these characteristics.

Sampling Frame and Population

Population: Our target population is the tenured and tenure-track faculty (Assistant, Associate and Full Professors, including named chairs) whose **primary appointment** is in one of Trinity's 27 academic departments. We settled on this definition because:

1. It matches our questions, which concern active regular-rank faculty (ex. including emeriti, for example, would inflate years since Ph.D.);
2. It lets every person be counted exactly once, so that $\pi_i = n_h/N_h$ is each person's true chance of selection

Eligibility rules: A person is eligible if the title of their primary appointment begins with "Assistant Professor," "Associate Professor" or "Professor." We excluded non-tenure-track titles:

Excluded group	Example
Emeriti	Professor Emeritus
Professors of the Practice	Associate Professor of the Practice of Dance
Lecturers and instructors	Lecturing Fellow, Instructor
Research, adjunct, visiting and clinical faculty	Assistant Research Professor, Visiting Professor

Departments:

We took the 27 departments from Trinity's own division pages: 12 in the Humanities, 8 in the Natural Sciences and 7 in the Social Sciences. Interdisciplinary programs (e.g., Dance, Writing & Rhetoric, Education, Linguistics, ROTC) and affiliated centers were not counted as departments. This loses no one because all 104 primary appointments in the excluded programs are professors of the practice, lecturers, instructors, or visiting, adjunct or emeritus faculty (do not meet our eligibility rules)

Multiple appointments:

Many faculty hold joint appointments, so if a person were listed in each of their departments, their real chance of selection would exceed the probability our weights assume, which would bias the estimates. We therefore counted each person once, in their primary department as recorded by Scholars@Duke. We decided to exclude faculty whose primary appointment is outside Trinity, even if they hold a secondary Trinity appointment.

Building the Sampling frame:

1. The starting list (The Scholars@Duke people search filtered to *Trinity College of Arts & Sciences* and *Faculty*) returns 1,440 people. Using a browser script, we recorded only each person's name, title and profile link to filter down to our sampling frame of 495 eligible faculty. Most are outside our population: 306 emeriti, 217 lecturers and instructors, 111 professors of the practice, 104 research, adjunct or visiting faculty, and 207 people whose primary appointment is outside Trinity's departments.
2. Our script, build_frame.py, makes one request to the Scholars@Duke public data feed for Trinity, which labels every appointment as primary or secondary. It keeps primary appointments in the 27 departments that have tenure-track titles, and writes the frame to **roster_all.csv.** It saves only frame fields (name, title, department, rank, profile link).
3. Checking against department websites. As per the assignment description, we compared the frame with the faculty pages of all 27 department websites. 493 of 495 frame members appear on their own department's page. The 2 exceptions are Physics professors who are listed as primary Physics faculty on Scholars@Duke but are missing from the Physics website. The websites also list some people who are not in our frame. All of them are excluded by our rules; every difference between the websites and our frame is therefore explained by our eligibility rules. Using Scholars@Duke to assign each person a single primary department simply removes the duplicates that the websites would create.

Final frame: N = 495 faculty in 27 departments (Humanities 139, Natural Sciences 208, Social Sciences 148). Department sizes N_h range from 3 (Slavic & Eurasian Studies) to 47 (Economics).

Population-wide outcome data: For non-sampled faculty we recorded only frame information (name, title, department), the same information shown on public department pages. Outcome data (sex, grants, education) were collected only for the 124 sampled faculty, after the sample was drawn.

Sampling Design

After getting the sampling frame, we decided to implement stratified random sampling with approximately proportional allocation to improve the precision of our estimates. We expect faculty within the same department to have relatively similar characteristics, such as the

proportion of women and grant funding. Thus, by taking faculty from every single department, we are able to get a better representation of the entire population, which could help reduce the variance of our estimates compared to simple random and cluster sampling. If we took a simple random sample, there is a possibility we wouldn't sample faculty from smaller departments, potentially increasing the variance of our estimates. If we took a cluster sample, we would be sampling entire departments whose faculty tend to have similar characteristics. By not including other departments, we would be providing less information than sampling individuals across all departments, which could increase the variance of our estimates. Furthermore, although cluster sampling might have been more convenient for data collection, scraping allowed us to implement stratified sampling without substantially increasing the time required for data collection. Thus, we thought the best method was to do stratified sampling.

Approximately proportional allocation made it straightforward to determine our sample sizes. We wanted to get around a 25% sampling proportion, with a minimum of two faculty per department, rounding to the nearest whole number. Thus, we initially targeted a sample of 120, as it is roughly 25% of 495. Given N_h is the population size in stratum h, our $n_h \approx 120\frac{N_h}{495}$. We ended with a final sample of 124, as shown in our faculty.csv dataset. We then randomly sampled n_h faculty without replacement from the N_h faculty in each department. We were then able to get our selection probability and weight using the equations $\pi_i = \frac{n_h}{N_h}$ and $w_i = \frac{N_h}{n_h}$. Our faculty.csv file shows each department's population size, sample size, and sampling weight.

Data Collection

Process:

sample.R writes the sample to sample.csv. collect.py then looks up each of the 124 sampled faculty, making two requests 1.5 seconds apart: one to their public profile page, to find their Scholars ID, and one to their record in the Scholars@Duke public data feed. These records hold the education and grant information shown on the profile's *Credentials* and *Research* tabs. The script codes the education and grant variables and writes to data.csv. We then coded sex by hand and saved the final dataset as **faculty.csv**, which contains name, department, N_h, n_h, π_i, w_i and the five outcome variables.

Definition of a grant:

We define a grant as any funding award listed in a faculty member's Scholars@Duke record: research grants, fellowships and gifts. Scholars@Duke groups these together on profiles under "Selected Grants" and "Fellowships, Gifts & Sponsored Research." We included all three because each provides funding for a faculty member's work. Counting only formal research grants would undercount faculty in the humanities and arts, where support more often comes as fellowships or foundation gifts. A grant is ongoing in 2026 if its period, from start date to end

date, includes any part of 2026: it begins in or before 2026 and ends in or after 2026. We took start and end dates from the Scholars@Duke data feed, while the profile page shows years only. A faculty member is coded grant_2026 = 1 if they have at least one such grant.

Coding rules:

Variable	Rule	Result (n = 124)
sex	1 = female, 0 = male, coded by hand	53 female, 71 male
grant_2026	1 if any listed grant, fellowship or gift runs at some point in 2026; 0 otherwise, including when none are listed	58 coded 1
has_phd	1 if a Ph.D. is listed; 0 if education is listed without one; missing if no education is listed	117 / 2 / 5
years_since_phd	2026 minus the Ph.D. year (Ph.D. holders only)	1 to 55 years
phd_duke	1 if the Ph.D. is from Duke University (Ph.D. holders only)	5 of 117

Ambiguous and missing cases:

- **No education listed (5 faculty):** Ph.D. status is unknown, so we left them out of the Ph.D. subpopulation for Q3 and Q4. They remain in Q1 and Q2. This assumes they resemble the Ph.D. holders whose education is listed (missing at random). With only 5 of 124 affected, the impact should be small.
- **Non-Ph.D. doctorates (2):** A Doctor of Musical Arts and a Norwegian *Dr. Art.* were coded has_phd = 0, because the question asks specifically about Ph.D.s.
- **"D.Phil." (3):** Three faculty list a D.Phil., the name some universities (e.g., Oxford) use for the Ph.D. We counted these as Ph.D.s. (All three are from U.S. universities that award the Ph.D., so the label is likely a recording quirk in Scholars@Duke.)
- **Two Ph.D.s (1):** We used the earlier one.
- **No grants listed (36).** Coded grant_2026 = 0.
- **Awards with no end date (20 faculty):** Treated as lasting only their start year, since most are one-time fellowships or gifts. This is the one rule that can materially affect a result: treating these awards as still ongoing would change 10 faculty to 1 and raise the Q2 estimate; our Q2 estimate is therefore conservative.

Statistical Analysis and Results

We used the survey package in R to implement the 27-stratum design. We were able to get the point estimates, standard errors, and 95% confidence intervals for each question. We were able to use the previously calculated sampling weights and department population sizes N_h to account for finite population corrections.

For the last two questions, we conducted a subpopulation analysis using the original survey design. More specifically, we filtered the original survey design so that it didn't include individuals without a Ph.D. degree and faculty that had missing information.

We obtained these results:

Goal	**Estimate**	**SE**	**95% CI**
Women	42.38%	3.83%	(34.86%, 49.89%)
Grant in 2026	46.79%	2.88%	(41.14%, 52.43%)
Years since Ph.D.	23.46 years	0.94 years	(21.62, 25.30)
Duke Ph.D.	4.48%	1.72%	(1.11%, 7.86%)

Interpretations:

1. We estimate the fraction of faculty who are women is 42.38% with a SE of 3.83%. We have a 95% CI of (34.86%, 49.89%) for this estimate. Since the interval is entirely below 50%, the results suggest that women make up less than half of eligible faculty.
2. We estimate the fraction of faculty who have at least one grant that is ongoing in 2026 is 46.79% with a SE of 2.88%. We have a 95% CI of (41.14%, 52.43%) for this estimate. The results suggest that around half of Duke's faculty we consider to be eligible has at least one ongoing grant in 2026.
3. Among faculty who have a Ph.D. degree, we estimate the average number of years since they received the Ph.D. degree is 23.46 years with a SE of 0.94 years. We have a 95% CI of (21.62, 25.30) years for this estimate. This suggests that, on average, Duke's eligible faculty with Ph.D. degrees received them approximately 23 years ago.
4. Among faculty who have a Ph.D. degree, we estimate the fraction who received their Ph.D. from Duke is 4.48% with a SE of 1.72%. We have a 95% CI of (1.11%, 7.86%) for this estimate. The results suggest that most eligible faculty with Ph.D. degrees received them from other universities.

Potential Limitations:

One possible limitation of our study is that we rely heavily on the accuracy and completeness of information collected from Scholars@Duke. The public information may not include every eligible faculty member in Trinity College, and some profiles may be outdated, causing us to exclude eligible faculty or include people not meeting our criteria. Furthermore, some faculty may have ongoing grants that are not listed on their profiles, causing us to underestimate the fraction of faculty with current grants. Finally, we manually determined sex using profile information, which could lead to classification errors.

Additionally, some faculty profiles on the website did not provide complete information about their Ph.D. degrees. For these individuals, we could not determine when or where they received their Ph.D., so we excluded them from the Ph.D.-related analyses. Excluding these observations from the related analyses (goals 3 and 4) could introduce bias if these faculty systemically differ from those with complete information. For example, faculty who don't have information about educational background may be less likely to have a Ph.D. degree. In the scenario, the faculty included in the analyses would not fully represent the actual population of Ph.D. holders.

Conclusion

By studying 124 faculty profiles, only about a quarter of the population, we were able to produce estimates for all 495 tenure-track faculty in Trinity College. That is what makes probability sampling powerful – because every faculty member has a known, nonzero chance of selection, the randomness of the draw becomes something we can quantify rather than a source of bias. Stratifying by department ensured that every corner of Trinity was represented, and the weights then let each sampled professor stand in for the colleagues they represent.

The project also showed that the hardest part of a survey often comes before any data is even collected; that is, deciding exactly who belongs in the population. For example, joint appointments, professors of the practice and emeriti all blur the line of who counts as a "Trinity faculty member." How carefully we drew that line mattered as much as the formulas we applied afterward.

The results give a picture of Trinity's faculty. We estimate that about 42% are women (95% CI: 35% to 50%), so women remain a minority. Roughly half hold a grant ongoing in 2026. Our 47% estimate is conservative and could be closer to 55% under a more generous reading of the grant records. The average faculty member received their Ph.D. about 23 years ago, which indicates a more senior faculty. Lastly, only about 4.5% earned their Ph.D. at Duke, consistent with the academic norm of hiring scholars trained elsewhere rather than one's own graduates.

Survey sampling is often taught as formulas, but this project clearly demonstrates the power it has for practical purposes. With a well-built frame and a random sample, a few hours of careful work can answer questions about an entire institution.